Foundations of computational finite element methods

August–November Semester, 2024–2025

Instructor: Dr Gopikrishnan C R
Office: 10, D03-FF (Dr. APJ Abdul Kalam Block)
Venue: C06 - 105
Time: Moday, Wednesday, Friday (3.00 – 4.30 PM)

Syllabus

- Weak formulation of elliptic and parabolic problems, dissertation in general finite dimensional spaces, degrees of freedom, matrix formulation and invertibility, boundary conditions (Dirichlet, Neumann, Robin)
- Conforming Lagrange P^1 finite element method
  - Poisson equation and two point boundary value problems in 1D, implementation of refinements and error subroutines
  - Poisson equation with Dirichlet, Neumann, and Robin boundary conditions in 2D, refinement strategies, L^2, L^∞, and H^1 error subroutines
  - Discretisation of parabolic problems
  - Visualisation using ParaView and Matlab functionalities, animation procedures in ParaView.
- Non-conforming finite element methods, Crouziex–Raviart and Raviart–Thomas FEM.
- Discontinuous Galerkin methods, interior penalty methods (symmetric and non-symmetric interior), global element method.
- Discretisation of nonlinear problems through Newtons and fixed point method, relaxation techniques.
- Finite volume methods, upwind method with Godunov flux, flux limiters, MUSCL schemes.

Note 1. No references are listed for this course. Lecture notes will be provided.

Note 2. As this is an informal course, there is no compulsory attendance or assessment. However, participants are encouraged to attend the course regularly to develop the ability to work seamlessly with FEM codes.

Note 3. After each unit, exercises are provided for practise. Participants are strongly advised to complete the exercises. Submissions are encouraged, not mandatory.

Note 4. The codes will be published in the GitHub repository Computational-Course-FEM and on the course website.